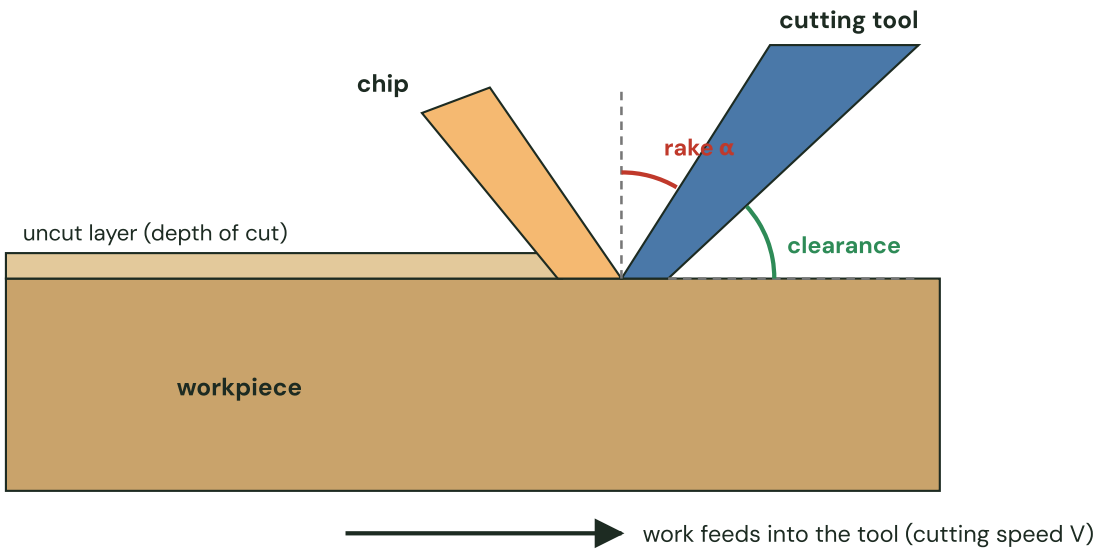
UNIT 8 · TOPIC 4(A)

Metal Cutting, Machining and Tool Life

Syllabus topic: Metal cutting and machining; jigs, fixtures and gauges

Chip formation and tool geometry



Orthogonal metal cutting — rake and clearance angles (redrawn from the source figure)

- Machining removes material as a CHIP when a wedge-shaped tool is forced through the work.
- **RAKE ANGLE** (positive/negative) controls chip flow and cutting force — higher rake → lower force but a weaker edge.
- **CLEARANCE (relief) angle** stops the tool flank rubbing the work.
- Cutting fluids cool and lubricate, extending tool life and improving finish.

Chip type	When
Continuous	Ductile metals
Continuous with built-up edge (BUE)	Ductile metals at certain speeds
Discontinuous	Brittle metals

Machining processes

Process	Produces
TURNING (lathe); facing; parting	Cylindrical parts — external/internal
DRILLING	Holes
BORING	Enlarges/finishes existing holes
REAMING	Finishes holes
MILLING (up vs down milling)	Flat/contour surfaces, slots
SHAPING / PLANING	Flat surfaces
GRINDING	Abrasive finishing; hard materials
Sawing, broaching	—
Modern: CNC, EDM, laser/plasma/water-jet cutting	—

Cutting parameters: speed (m/min), feed (mm/rev) and depth of cut.

Tool materials and tool life

In ascending performance: high-carbon steel → HIGH-SPEED STEEL (HSS) → cemented CARBIDE → CERAMICS/cermets → CBN/diamond.

Taylor's tool-life equation: $V \cdot T^n = C$ (V = cutting speed, T = tool life, C = constant)

Tool material	Exponent n (source)
HSS	≈ 0.16
Uncoated carbide	≈ 0.30
Coated carbide	≈ 0.35
Ceramic	≈ 0.40

- A larger n means tool life is LESS sensitive to speed.
- **Wear modes:** flank wear (VB), crater wear, edge chipping.

Textbook ranges differ (e.g., Kalpakjian: HSS 0.08–0.2, carbides 0.2–0.5, coated carbides 0.4–0.6, ceramics 0.5–0.7). The ORDER HSS < carbide < ceramic is what exams test.

Worked example. n = 0.2; a tool lasts T_1 = 60 min at V_1 = 40 m/min. At V_2 = 60 m/min:
$T_2 = T_1 \cdot (V_1/V_2)^{(1/n)} = 60 \times (40/60)^5 = 60 \times 0.1317 \approx$ **7.9 min** — a 50% speed rise cuts tool life to about one-eighth.

Practice MCQs

Q47. Machining removes material in the form of a:

(a) Weld bead
(b) Chip
(c) Casting
(d) Powder only

Q48. Increasing the rake angle of a cutting tool generally:

(a) Increases cutting force
(b) Decreases cutting force but weakens the edge
(c) No effect
(d) Stops chip flow

Q49. The clearance (relief) angle prevents:

(a) Chip curling
(b) The tool flank rubbing the workpiece
(c) Overheating only
(d) Vibration

Q50. A built-up edge (BUE) tends to form when machining:

(a) Brittle metals
(b) Ductile metals at certain speeds
(c) Ceramics
(d) Glass

Q51. Cylindrical parts are produced mainly by:

(a) Milling
(b) Turning (lathe)
(c) Grinding only
(d) Shaping

Q52. Enlarging and finishing an existing hole is:

(a) Drilling
(b) Boring
(c) Blanking
(d) Knurling

Q53. Flat surfaces and slots are commonly produced by:

(a) Turning
(b) Milling
(c) Reaming
(d) Drawing

Q54. The three basic cutting parameters are:

(a) Speed, feed, depth of cut
(b) Voltage, current, time
(c) Load, area, stress
(d) Mass, velocity, force

Q55. In ascending order of cutting performance the tool materials are:

(a) Carbide < HSS < steel
(b) HSS < carbide < ceramic < CBN/diamond
(c) Diamond < HSS
(d) Ceramic < steel

Q56. Taylor's tool life equation is:

(a) $V \cdot T^n = C$
(b) $V + T = C$
(c) $V/T = C$
(d) $V \cdot T = n$

Q57. In $V \cdot T^n = C$, V and T are:

(a) Volume and torque
(b) Cutting speed and tool life
(c) Voltage and time
(d) Velocity and temperature

Q58. The exponent n for HSS tools is about:

(a) 0.16
(b) 0.30
(c) 0.40
(d) 0.60

Q59. The exponent n for ceramic tools is about:

(a) 0.10
(b) 0.16
(c) 0.30
(d) 0.40

Q60. A higher value of n means tool life is:

(a) More sensitive to speed
(b) Less sensitive to cutting speed
(c) Zero
(d) Infinite

Q61. With n = 0.2, raising speed from 40 to 60 m/min changes tool life from 60 min to about:

(a) 40 min
(b) 20 min
(c) 8 min
(d) 60 min

Q62. Common tool wear modes are:

(a) Flank and crater wear
(b) Rust only
(c) Melting
(d) Corrosion only

Answers

Q47 — (b) Chip
Q48 — (b) Decreases cutting force but weakens the edge
Q49 — (b) The tool flank rubbing the workpiece
Q50 — (b) Ductile metals at certain speeds
Q51 — (b) Turning (lathe)
Q52 — (b) Boring
Q53 — (b) Milling
Q54 — (a) Speed, feed, depth of cut
Q55 — (b) HSS < carbide < ceramic < CBN/diamond
Q56 — (a) $V \cdot T^n = C$ — *$V \cdot T^n = C$.*
Q57 — (b) Cutting speed and tool life
Q58 — (a) 0.16 — *n(HSS) ≈ 0.16 in the source (textbook ranges vary; the order is what is tested).*
Q59 — (d) 0.40 — *n(ceramic) ≈ 0.40 in the source.*
Q60 — (b) Less sensitive to cutting speed — *Higher n = life less speed-sensitive.*
Q61 — (c) 8 min — *$T_2 = 60 \cdot (40/60)^{(1/0.2)} = 60 \cdot (0.667)^5 \approx 7.9$ min.*
Q62 — (a) Flank and crater wear

Sources: Hajra Choudhury — Elements of Workshop Technology (Vols I & II); Kalpakjian — Manufacturing Engineering & Technology; standard production-engineering texts; NPTEL manufacturing lectures.